

Machine Learning

Instructor: Hafiz Abdul Rehman

Lecture 01: Introduction

About Me

- MS (CS)
 - COMSAT University Islamabad (Lahore Campus)
 - 2020-2022
- Research Officer
 - AL-Khwarizmi Institute of Computer Science (KICS) UET Lahore
 - 2019-2023
- Lecturer
 - Computer Science Department, SST

About Me

- Instructor: Hafiz Abdul Rehman
- Email: hafiz.rehman@umt.edu.pk
- Office: Room No 404, STD
- Google Scholar Profile: [Hafiz Abdul Rehman](#)
- GitHub Profile: [Hafiz Abdul Rehman](#)
- LinkedIn Profile: <https://www.linkedin.com/in/ha-rehman/>

About Me : Research Profile



Hafiz Abdul Rehman 

 FOLLOWING

Research Officer KICS UET Lahore

Verified email at kics.edu.pk

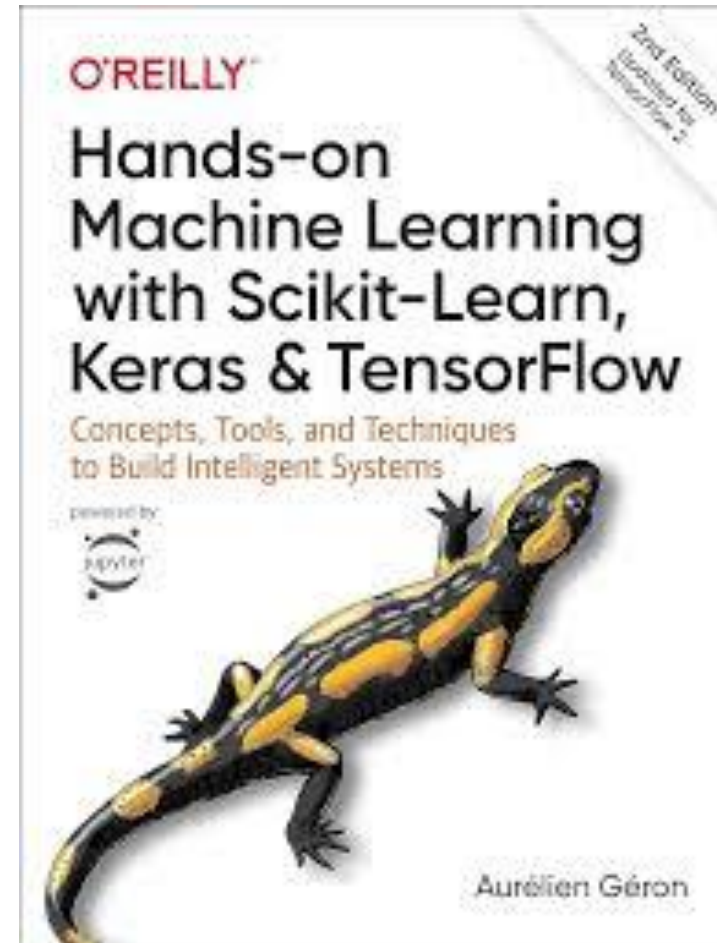
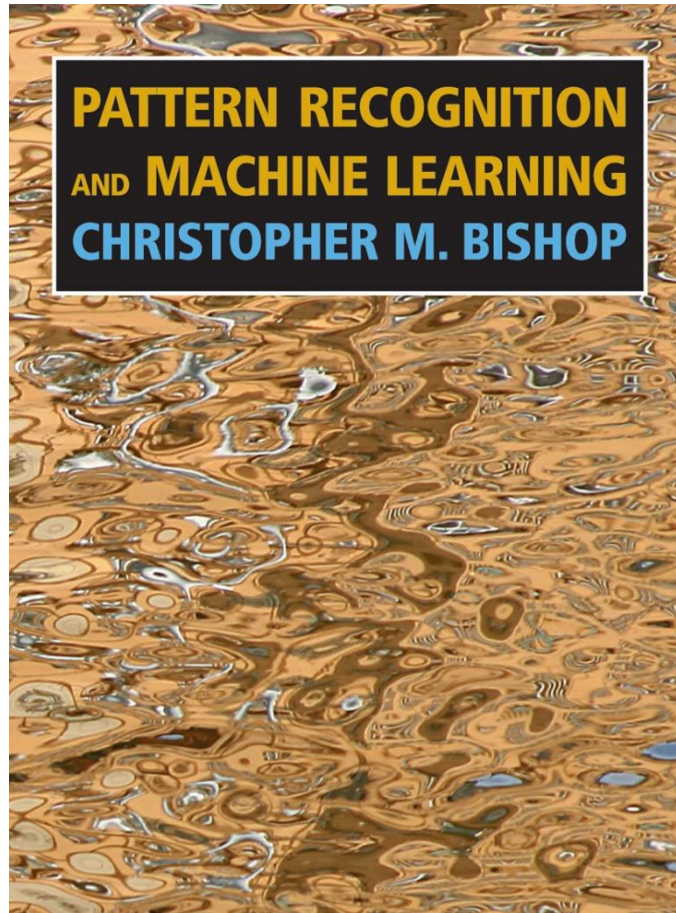
Machine Learning Deep Learning Computer Vision Natural Language Processing
Genome Sequencing

☐	TITLE			CITED BY	YEAR
☐	Comprehensive comparative genomic and microsatellite analysis of SARS, MERS, BAT-SARS, and COVID-19 coronaviruses			11	2021
	HA Rehman, F Ramzan, Z Basharat, M Shakeel, MUG Khan, IA Khan Journal of Medical Virology 93 (7), 4382-4391				
☐	DeepFireNet-A Light-Weight Neural Network for Fire-Smoke Detection			8	2021
	M Mubeen, MA Arshed, HA Rehman International Conference on Intelligent Technologies and Applications, 171-181				
☐	A 16× 16 Patch-Based Deep Learning Model for the Early Prognosis of Monkeypox from Skin Color Images			3	2024
	MA Arshed, HA Rehman, S Ahmed, C Dewi, HJ Christanto Computation 12 (2), 33				
☐	Leveraging coverless image steganography to hide secret information by generating anime characters using GAN			1	2024
	HA Rehman, UI Bajwa, RH Raza, S Alfarhood, M Safran, F Zhang Expert Systems with Applications 248, 123420				
☐	Enhancing Weather Scene Identification Using Vision Transformer				2024
	C Dewi, MA Arshed, HJ Christanto, HA Rehman, A Muneer, S Mumtaz World Electric Vehicle Journal 15 (8), 373				

Course Details

- Course Code: CS446
- Assessment
 - 20% Coursework
 - 30% Midterm,
 - 30% Final
 - 20% Project
- Office Hours: 11:00am – 1:00pm or When Door Open

Textbooks - ML



Pre Requisites - ML

Python Programming

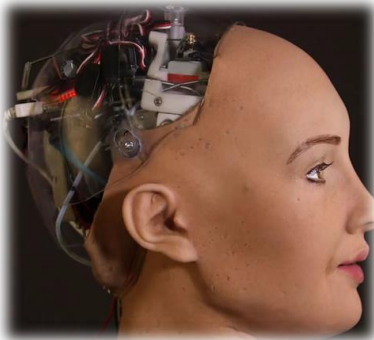
Linear Algebra | Calculus

Probability and Statistics

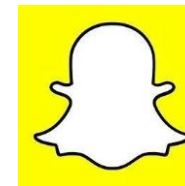
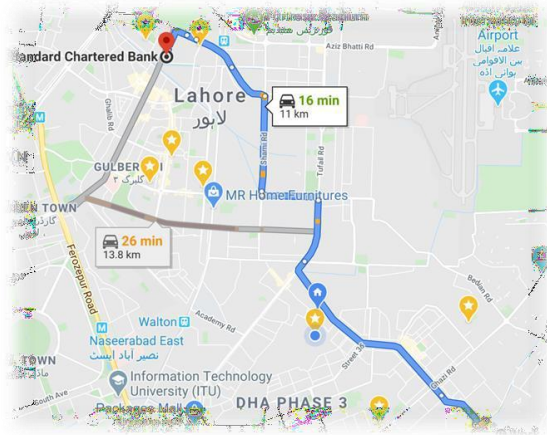
Course Outline

- Introduction to Machine Learning
- Linear Model with implementation
- Logistic Regressor
- Evaluation Measure
- Overfitting/Underfitting
- k-Nearest Neighbor, Naive Bayes
- Support Vector Machines
- Ensemble Methods
- Perceptron

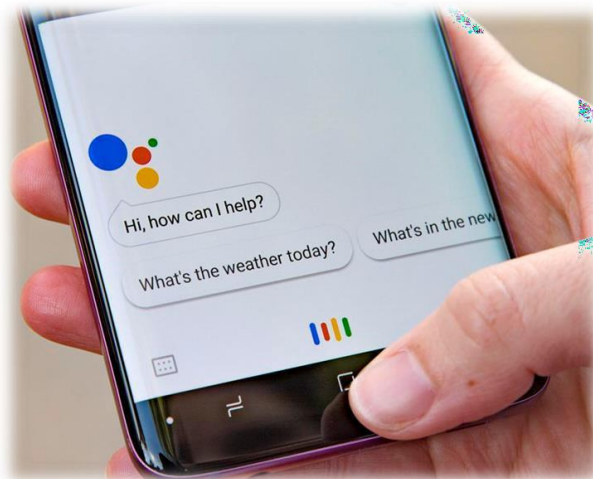
Machine Learning



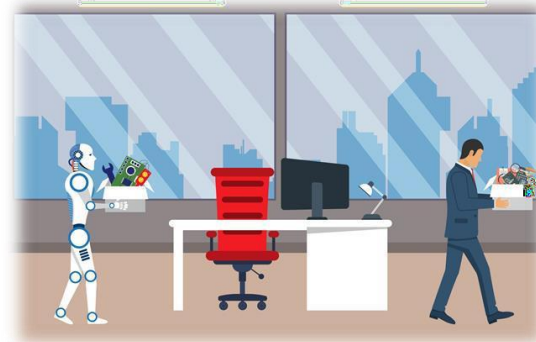
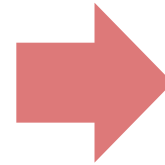
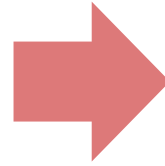
Machine Learning



Machine Learning



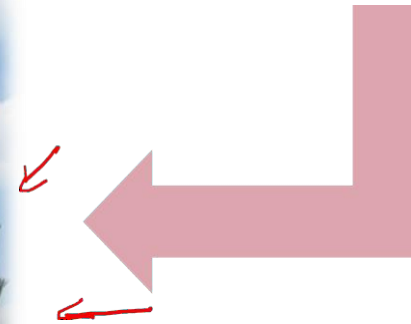
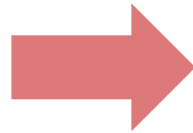
Machine Learning



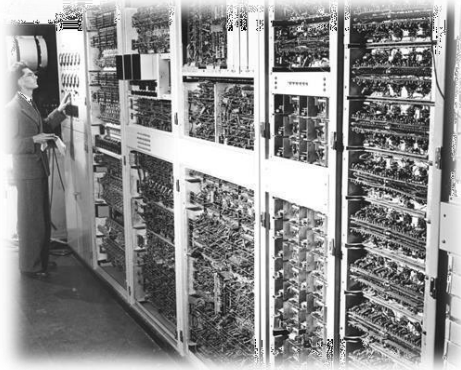
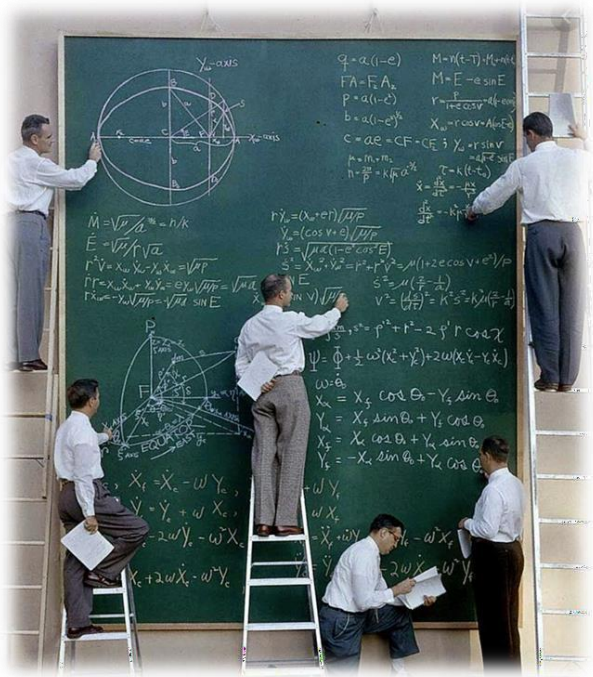
What is Machine Learning?

How does it work?

Machines as mechanical helpers



Machines as mechanical helpers



Machines as mechanical helpers



→ Cat

Is this a cat or a dog?

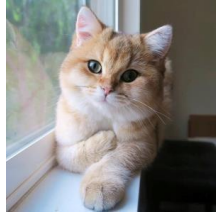


→ No

Should I hire this person?



Machines as mechanical helpers



{cat, dog}



→ cat



{happy, sad, angry,
surprised, neutral}



→ happy



{empty, full}



→ empty



Machines as mechanical helpers



{All words of the language}



Hospital



Return the *best* of {all
board positions one black
move from current}



How do we train a classifier?

How to train your intern?

- **How would you train a new intern to conduct job Interviews?**
- **Option 1: Teach all the complicated rules**
 - Grades are important
 - University is important
 - Great grades – Good university – All good!
 - Bad grades – Unknown university – Not so good
 - Bad grades – Good university – ?
 - Good grades – unknown university – ?
- Still there would be exceptions
- Murky thresholds and gray areas
- Very hard to instill intuitive and experiential knowledge

How to train your intern?

- **How would you train a new intern to conduct job Interviews?**
- **Option 2: Make them sit and watch as a expert conducts interviews**
 - Learning by experience
 - Eventually, patterns start emerging
 - Let the intern get the intuition on their own
- More experience – better learning
- More exposure (balanced cases) – better learning
- Caveat!
 - What if the expert has systematic flaws of judgement aka biases?
 - Conduct sessions with many experts
 - What is they all share biases and stereotypes?
 - Initially, your intern could only be as good as the experts

How to train your machine?

Allowing the machines to learn
on their own using prior
decisions of experts is known as
Machine Learning

How to train your machine?

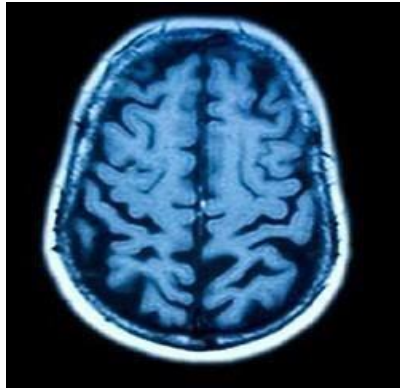
- The goal of AI was to make a machine more like a human
 - Give the machine a lot of world knowledge
 - A logical decision-making frame-work
- The Machine Learning framework seeks to make a better machine – not necessarily emulating a human
 - Based on Statistics and Optimization
 - Learn from labelled data
 - More data – more consistent decisions
 - More balanced data – more confident decisions
- **Certainty** in the real-world is a rare luxury – **Uncertainty** is the basis of ML that is quantified using probability and statistics

Traditional Computer Science

- Tasks like:
 - Play an audio/video file
 - Display a text file on screen
 - Perform a mathematical operation on two numbers
 - Sort an array of numbers using *Insertion Sort*
 - Search for a string in a text file



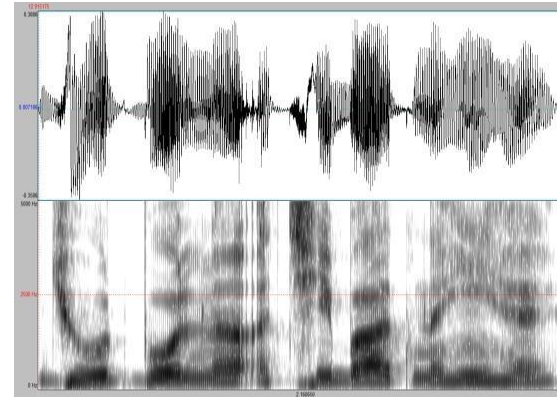
Traditional Computer Science



Tumor? y/n



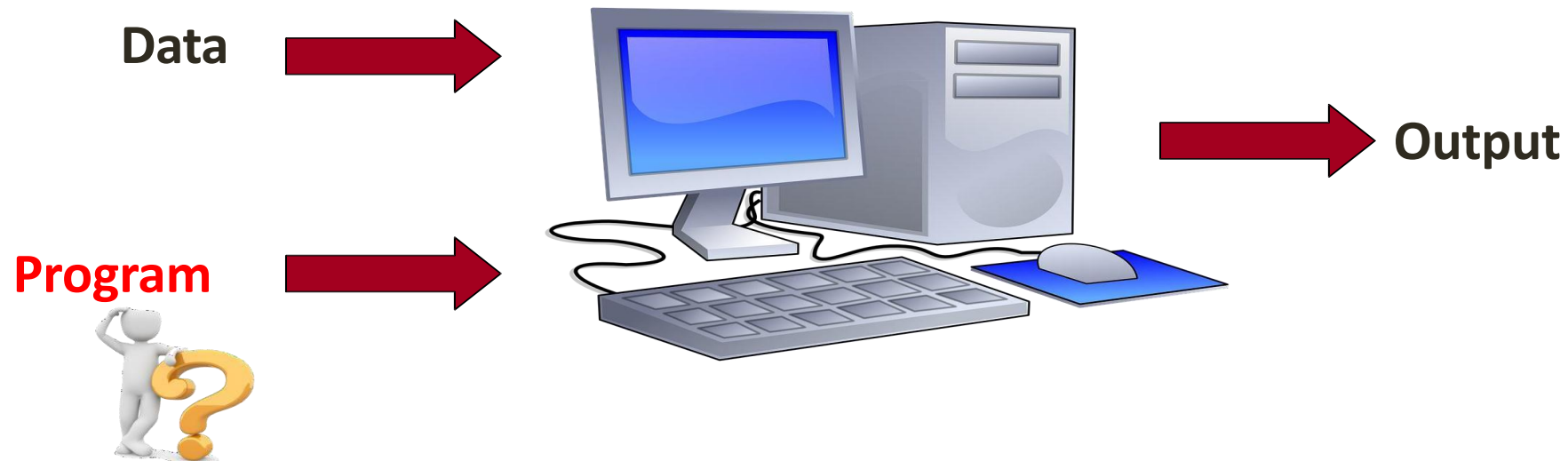
Price?



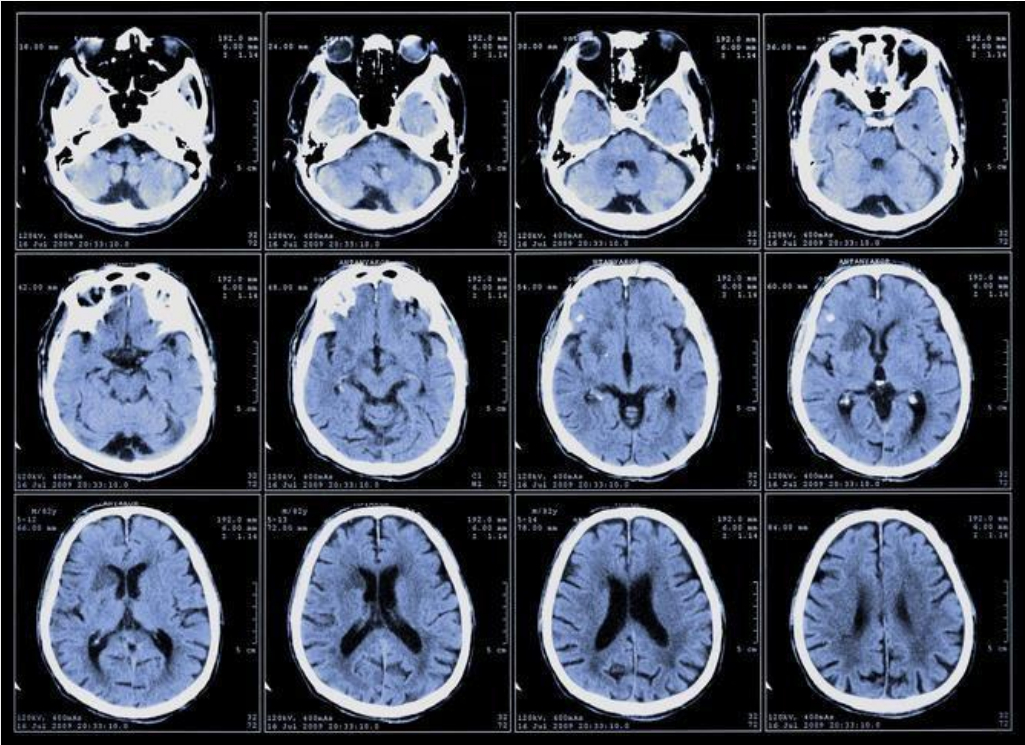
What was said?

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell) A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell) A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell) A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell) A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell) A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell)

Summarize
text



Past data



\$100,000



\$140,000



\$400,000



\$250,000



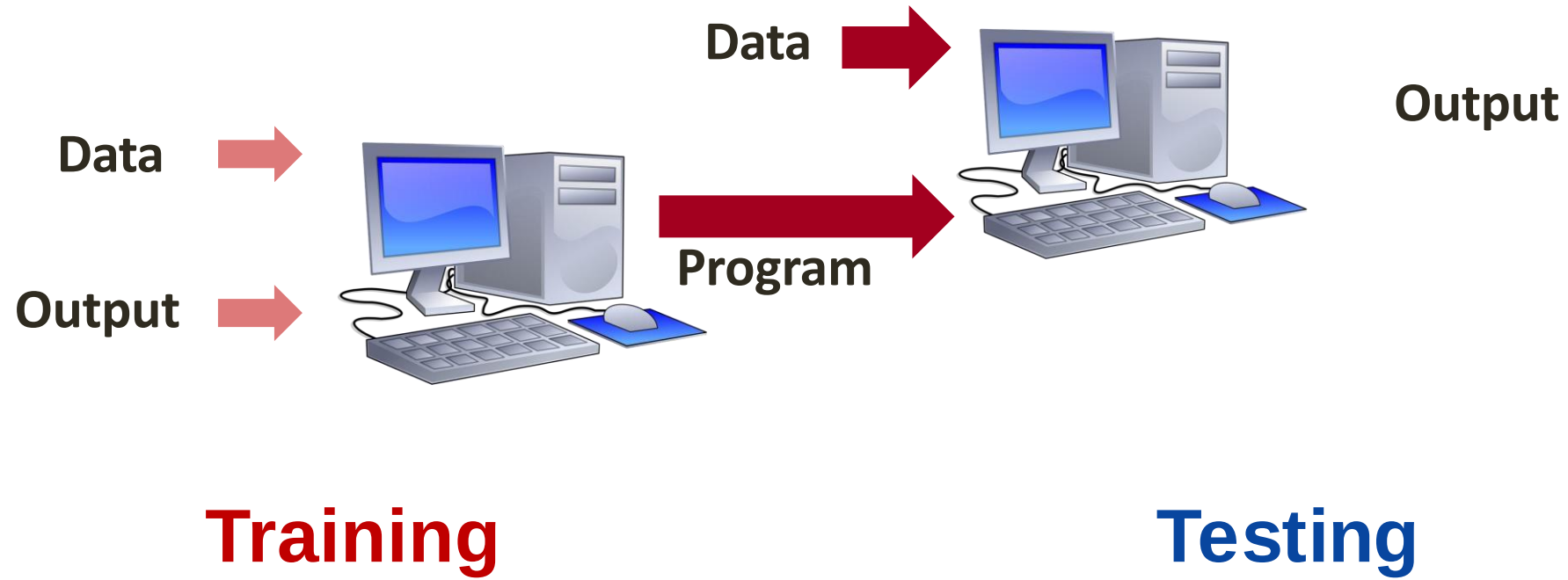
\$190,000



Traditional CS



Machine Learning



What is Machine Learning?

- **Formally:** A computer program **A** is said to learn from experience **E** with respect to some class of tasks **T** and performance measure **P** if its performance at tasks in **T**, as measured by **P**, improves with experience **E**. (Tom Mitchell, 1997)
- **Informally:** *Algorithms that improve on some task*
 - *with experience.*

Thank You

I wish for your better Career!